IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

Executive Summary

SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars; other providers cost upward of 165 million dollars each. Much of the savings is because SpaceX can reuse the first stage. Our goal was to predict the likelihood of a successful mission. If the first stages cannot be recovered the cost will be far greater.

Staff determined the key elements needed to build a database that would give a history of every successful and unsuccessful launch. Key factors like the launch date, location, target orbit, target landing site, booster version and reuse count, payload weight, special equipment (legs and gridfins), mission success, and landing success. This data was located on multiple web pages which we "webscraped" to build tables; other data was found simply by using web applications (API) to download the data. Staff then cleaned up the database fields ensuring codes were normalized and missing numbers replaced with statistically logical values. We then used this data in graphics and interactive charts to visualize the relationships between these values and a successful landing. Then, predictive software was used to train and test expanded data using multiple prediction methods for the most reliable results.

Results are very positive. We evaluated the factors listed above and determined SpaceX rapidly responded to mission failures with solid fixes to resolve known problems: booster models improved from V1.0 to Block 5, and solved payload weight and orbit target factors. Grid Fins were added to improve flight control during descent. Landing legs were improved to protect from hard landings. Launch time weather conditions are listed in the "Falcon User's Guide" by month and hour for prediction favorability percentage. (They still must work around Mother Nature!). Booster inventories and number of reuses continue to increase.

Introduction

Our project is the final course of the IBM Data Science series. In this project, we will use the methodology from data science to help examine factors that will affect sending rockets into space and return. SpaceX is succeeding in space because they figured out how to send rockets up and reuse them. They say it costs about 62 million dollars to launch their Falcon 9, which is 40% what other companies charge. They are reusing the first part of the rocket. But sometimes, SpaceX fails to recover that booster. That means costs are higher.

The goal of this project is to use machine learning to predict the viability of a successful landing . We must determine which factors correlate with success. Details such as Orbit Type, Payload Mass, Booster Version, Launch Site, Landing Site, and Weather Conditions must be evaluated.

Questions are: What causes landing failures? What is the life cycle of reused boosters? What geographical, environmental, and political issues face the Launch Sites?

Section 1

Methodology

Methodology

- Data collection methodology:
  - ➢ SpaceX Rest API
  - ➢ Web Scraping from Wikipedia page ‘List of Falcon 9 and Falcon Heavy Launches’
- Perform data wrangling
  - ➢ Transform or remove erroneous data, while appropriately handling missing values.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - ➢ Collected data was normalized, divided in training and test data sets and evaluated by four different classification models, with the accuracy of each model evaluated using different combinations of parameters.

Data Collection

The objective was to collect spacex launch data so that we can clean, analyze, visualize, and model data to predict a successful launch.

One of the datasets was collected by using Space X API and then the json response was normalized to a dataframe and then further preprocessing and analysis was done

Another dataset containing launch data was webscraped from a Falcon 9 Wikipedia page using the BeautifulSoup package.

Relevant data was extracted from HTML tables and put into a dataframe for further analysis

Data Collection – SpaceX API

Task 1: Request and parse the SpaceX launch data using the GET request

- Data elements are: Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial, Longitude, and Latitude
- For this project, remove records with a launch date greater than 2020-11-13
- API example: Get a json result for a record's Payload ID
  https://api.spacexdata.com/v4/payloads/5eb0e4b5b6c3bb0006eeb1e1

Task 2: Filter the dataframe to only include `Falcon 9` launches

- FlightNumber should be restarted with 1 for record 1 and increased sequentially

Task 3: Dealing with Missing Values

- Replace NaN fields in PayloadMass with the mean of the records
- Save the dataframe as a csv file for future study

Request SpaceX records as a static json response and create a "Base" DataFrame

Simplify "Base" DataFrame, keep key data elements and records before 2020-11-14

For Falcon 9 Heavy launches (three boosters) remove side boosters core and payload data

Call the API for each record to fill lists for the factors we want. Use id codes from the "Root" DataFrame columns to get clear text

Drop Falcon 1 records, restart FlightNumber

Calculate mean of PayloadMass and update NaN field using fillna(pl_mean)

GitHub URL of the completed SpaceX API calls

Data Collection - Scraping

TASK 1: Request the Falcon 9 Launch Wiki page by URL

- Use BeautifulSoup to scan the HTML-formatted text and create a structure that can be searched similar to json. There are a number of parsers to handle different versions and formatting of HTML.

TASK 2: Extract all column/variable names from the HTML table header

- A search will show the data is in the third table in the document and placed in TH elements . E,g: <th scope="col">Flight No. </th>
- Complex or badly formed HTML may cause unexpected characters and linefeed commands that will require strip and replace actions on strings. You may need to use a different parser in soup.

TASK 3: Create a data frame by parsing the launch HTML tables

- The table on the web uses two rows per record. The first row of each has the data. Each element in the row can be referenced as a list. Add the value to the appropriate dictionary

Request WikiPedia web page as an HTML formatted text stored in response.text

Analyze the structure of the page to determine how to identify the table, headings and columns

Retrieve data from <th> and <td> containers in each row of the table and add to dictionaries for each column name

Create a dataframe from the dictionaries and save as a csv

[GitHub URL of the completed WebScraping Notebook]()

Data Wrangling

- Create a dataframe using the csv file from the previous lesson.

TASK 1: Calculate the number of launches on each site

- Get a list of launch sites and count of launches

Launch Site

CCAFS SLC 40	KSC LC 39A	VAFB SLC 4E
55	22	13

TASK 2: Calculate the number and occurrence of each orbit

- Get a list of the name and count of each orbit in the column Orbit

Orbit

GTO	ISS	VLEO	PO	LEO	SSO	MEO	ES-L1	HEO	SO	GEO
27	21	14	9	7	5	3	1	1	1	1

- Note: GTO is a transfer orbit and not itself geostationary.

TASK 3: Calculate the number and occurence of orbit mission outcome

- Get a list of the mission outcome for each launch

Outcome

True ASDS	None None	True RTLS	False ASDS	True Ocean	False Ocean	None ASDS	False RTLS
41	19	14	6	5	2	2	1

TASK 4: Create a landing outcome label from Outcome column

- By using a field having 1 for success and 0 for failure in the Outcome category. Since this is 1 and 0, the mean (0.6666666) shows a 67 percent success rate.

[GitHub URL of the completed data wrangling Notebook]()

```
launch_site_counts =
df['LaunchSite'].value_counts()
print(launch_site_counts)
```

```
orbit_counts =
df['Orbit'].value_counts()
print(orbit_counts)
```

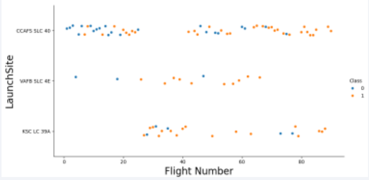
```
landing_outcomes =
df['Outcome'].value_counts()
print(landing_outcomes)
```

Create a new Class field with 1 for successful, 0 if not. By getting the mean of Class, you have the success percentage

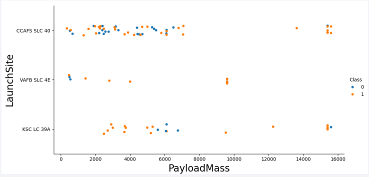
Save the dataframe as a csv

EDA with Data Visualization

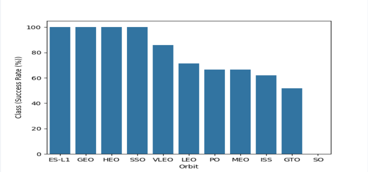
- ➢TASK 1: Visualize the relationship between Flight Number and Launch Site (Time Series)



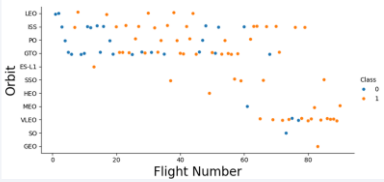
- ➢TASK 2: Visualize the relationship between Payload Mass and Launch Site (Comparison)



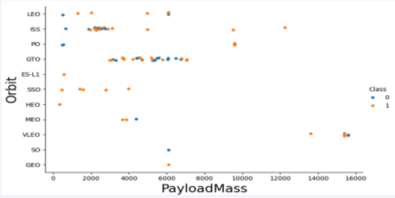
- ➢TASK 3: Visualize the relationship between success rate of each orbit type



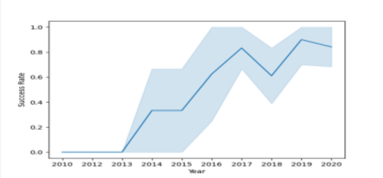
- ➢TASK 4: Visualize the relationship between FlightNumber and Orbit type



- ➢TASK 5: Visualize the relationship between Payload Mass and Orbit type



- ➢ TASK 6: Visualize the launch success yearly trend



- ➢TASK 7: Create dummy variables to categorical columns. Use the function get_dummies and features dataframe to apply OneHotEncoder to the columns Orbits, LaunchSite, LandingPad, and Serial. Assign the value to the variable features_one_hot.

- ➢TASK 8: Cast all numeric columns to float64. Now that our features_one_hot dataframe only contains numbers, cast the entire dataframe to variable type float64

- ➢We have compiled several charts that confirm the rapid improvement of safely landing stage one boosters on both landing pads and ocean drone ships.

GitHub URL of completed EDA with data visualization notebook

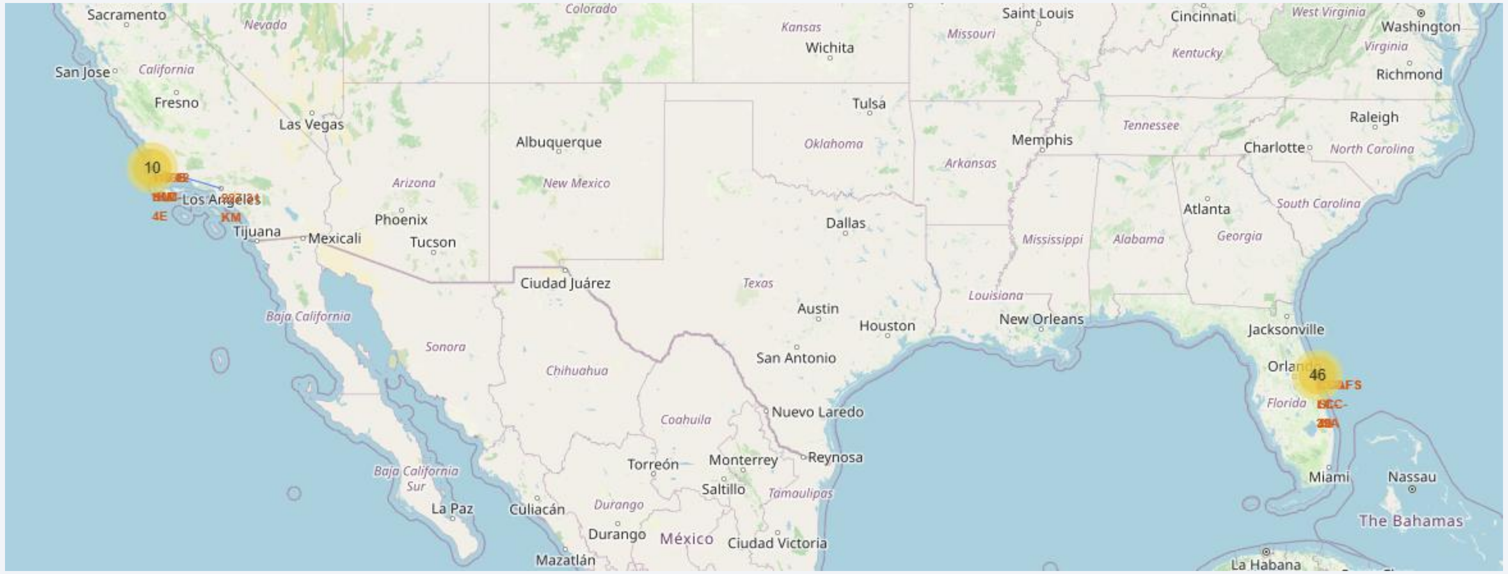
EDA with SQL

- Display the names of the unique launch sites mission
- Display 5 records where launch sites begin with the string 'CCA’
- Display the total payload mass carried by boosters launched by NASA (CRS)
- Display average payload mass carried by booster version F9 v1.1
- List the date when the first succesful landing outcome in ground pad was achieved
- List booster names which have success in drone ship and have payload mass greater than 4000 but less than 6000
- List the total number of successful and failure mission outcomes
- List the booster_versions that have carried maximum payload mass, use a subquery with a suitable aggregate fuction.
- List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.
- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

[GitHub URL of your completed EDA with SQL notebook]

Build an Interactive Map with Folium

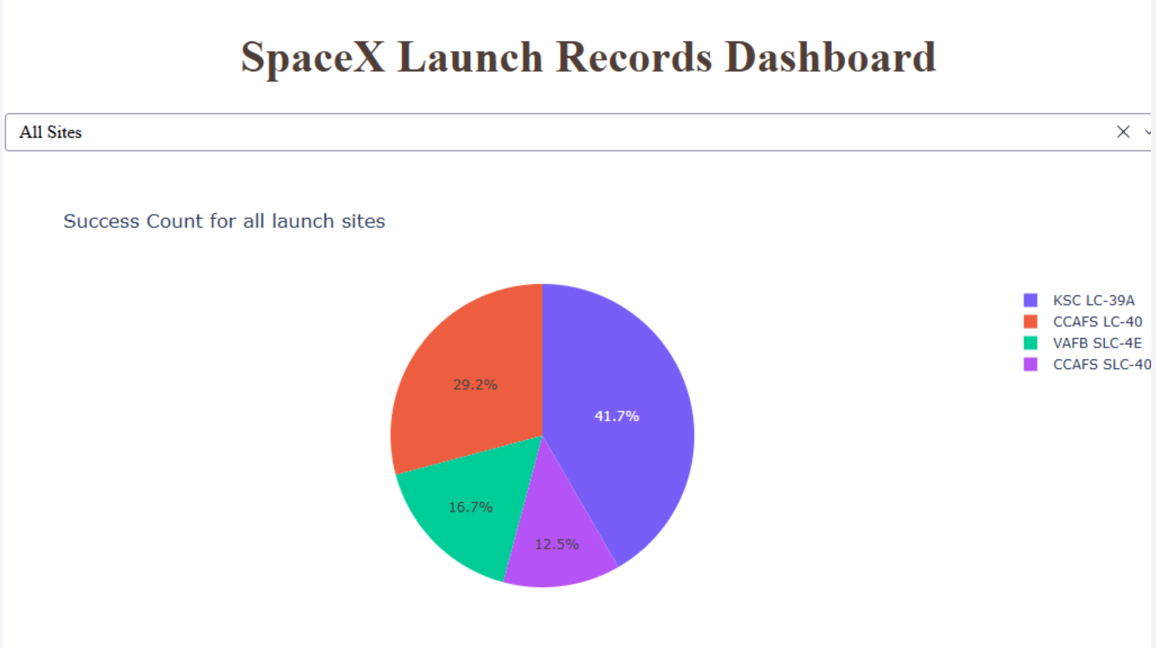
[View a completed usable folium map on your browser]   [Download the folium HTML file from GitHub]

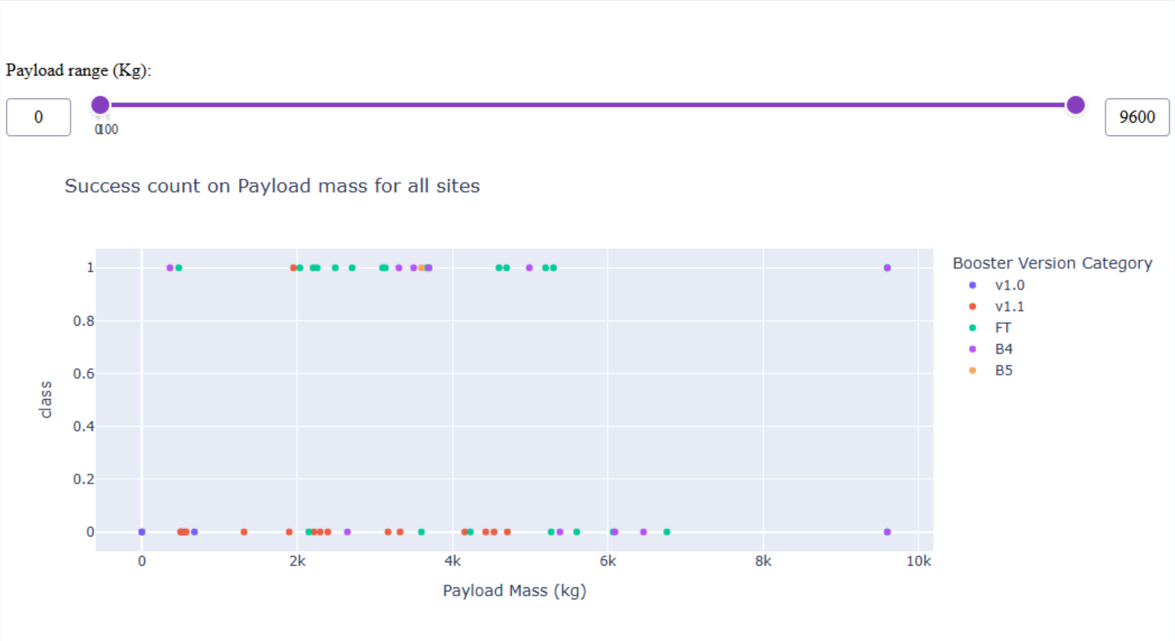


- Summary of map objects such as markers, circles, lines, etc. were created and added to a folium map
  - ➢ Circle: Add a highlighted circlular area with a text label on specific coordinates
  - ➢ Marker: Add a flagged position on a specific latitude and longitude
  - ➢ Marker Cluster: A container to hold multiple markers having the same coordinates for simplification until zoomed
  - ➢ Polyline: Draw a line from a site to the nearest coast, city, highway, etc. with distance displayed
  - ➢ MousePosition: The mouse position is displayed as longitude and latitude for easier gathering of needed coordinates

[GitHub URL of the completed interactive map with Folium Notebook]

Build a Dashboard with Plotly Dash



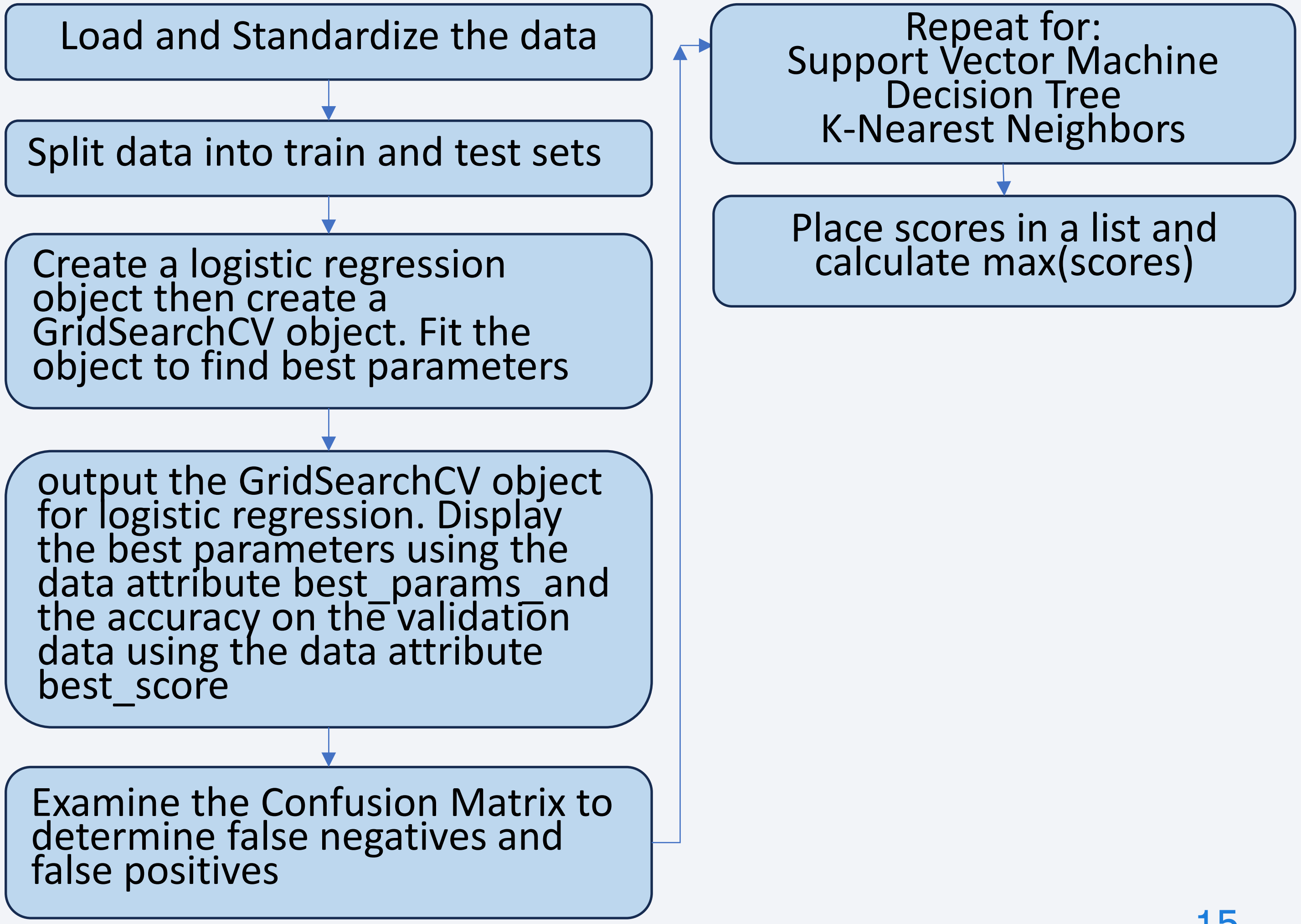


- A drop-down menu was added to the dashboard to allow users to choose the launch site(s) they wish to review
- An interactive pie chart was added to show users the total successful launch count for their selection from the drop-down menu
- A slider was added to allow users to choose the payload range
- A scatter plot was added to show users the correlation between their chosen payload and launch success for different sites

[GitHub URL of the completed Plotly Dash lab]() [View the Dashboard in your Browser]()

Predictive Analysis (Classification)

- LR, SVM, Decision Tree and KNN objects are created and fit with GridSearchCV object to find the best parameters, then the models are trained on the training set.
- The accuracy of test data is calculated for each machine learning model. The 4 methods LR, SVM, Decision Tree and KNN all achieved accuracy of 83.33%.



GitHub URL of the completed predictive analysis lab

Results

The GTO orbit is the busiest and most challenging for heavier weight payload. Launches should be made to the East (in Florida) to take advantage of the rotational speed of the Earth (1670Km/h). [1]

Data includes the "experimental" time period 2010 through 2016. In 2017 SpaceX ceased referring to launches as "Experimental". More concerning, every launch where the booster is declared "expendable" is treated as a failure. However, these were management decisions such as orbit, payload, separation speed, fuel burn, age of booster, removal of grid fins and landing legs to reduce weight. All boosters need to be retired (expended) eventually for safety or economy.[2] See page 20 for an example of "None None" (Expendable) to see the impact of this policy. Prior to Falcon 9 Block 5, all boosters were limited to only two flights; therefore they were expendable on the second flight. [3]

Weather impact has caused a number of failures and near-failures. SpaceX cannot accurately predict every flight but their user's guide provides the risk level for month and time of day [4] Page 24, Table 3.2

Garbage in, Gospel out. The data retrieved from API sources removes Falcon 9 Heavy side boosters and keeps only the core booster, which is more likely to be lost since it separates later at a higher speed. The two side boosters often made successful ground landings but were deleted from the database in Task 1: Request and parse the SpaceX launch data using the GET request in "Lab 1: Collecting the Data".

All four machine learning models performed the same in terms of prediction accuracy

Block 4 and Block 5 Boosters will increase the success rate of booster recovery

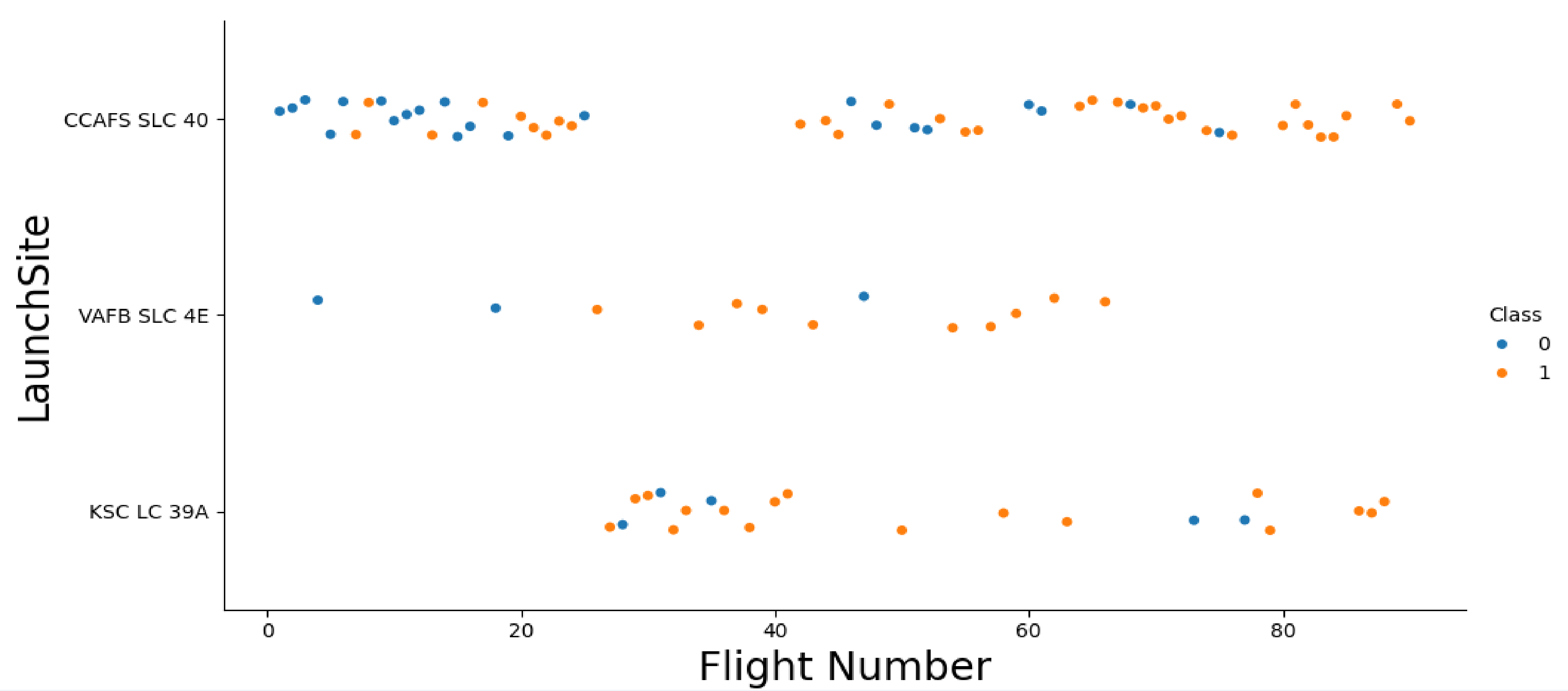
Reference

1) https://www.livescience.com/32721-why-are-rockets-launched-from-florida.html
2) https://en.wikipedia.org/wiki/SpaceX_reusable_launch_system_development_program
3) https://en.wikipedia.org/wiki/List_of_Falcon_9_first-stage_boosters
4) https://www.spacex.com/assets/media/falcon-users-guide-2025-05-09.pdf

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

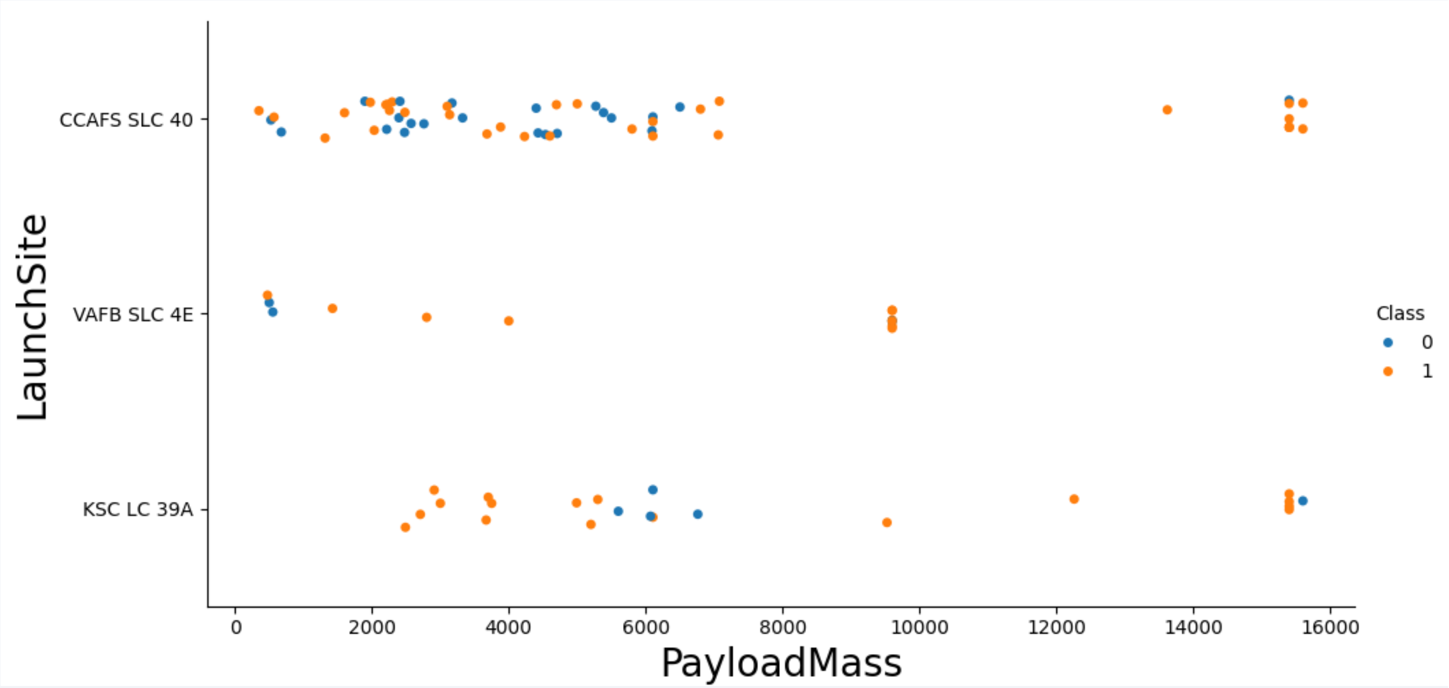


CCAFS SLC 40 was the primary test site. Launches became successful over time as boosters became more powerful and efficient and gridfins were added for better landing control. Landing legs were redesigned to protect from hard landings.

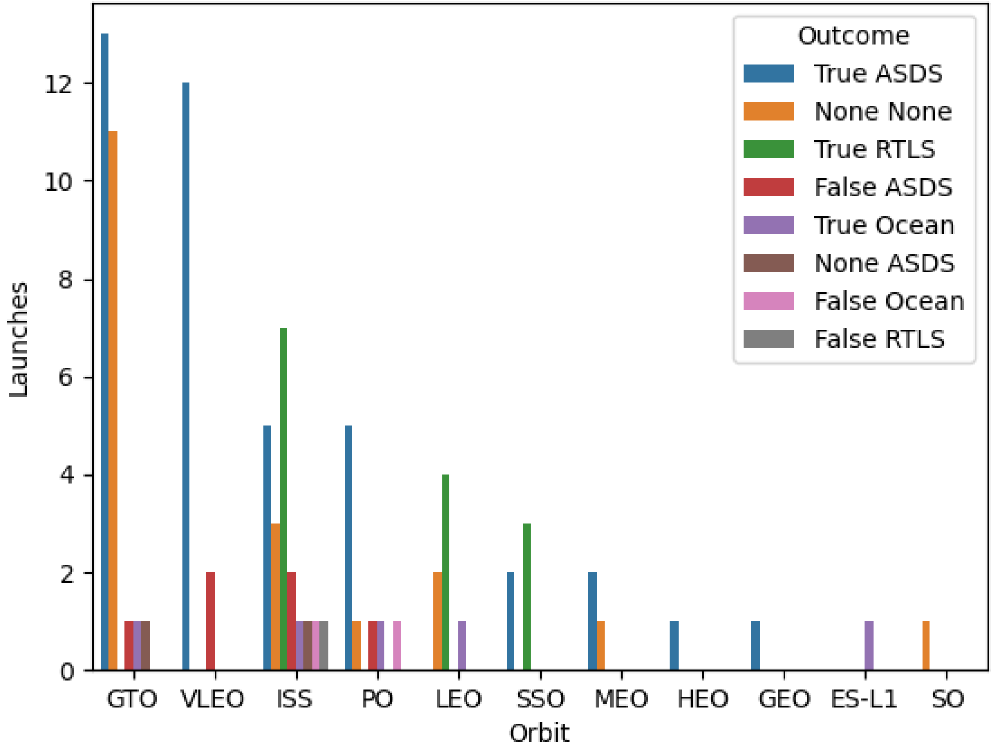
Payload vs. Launch Site

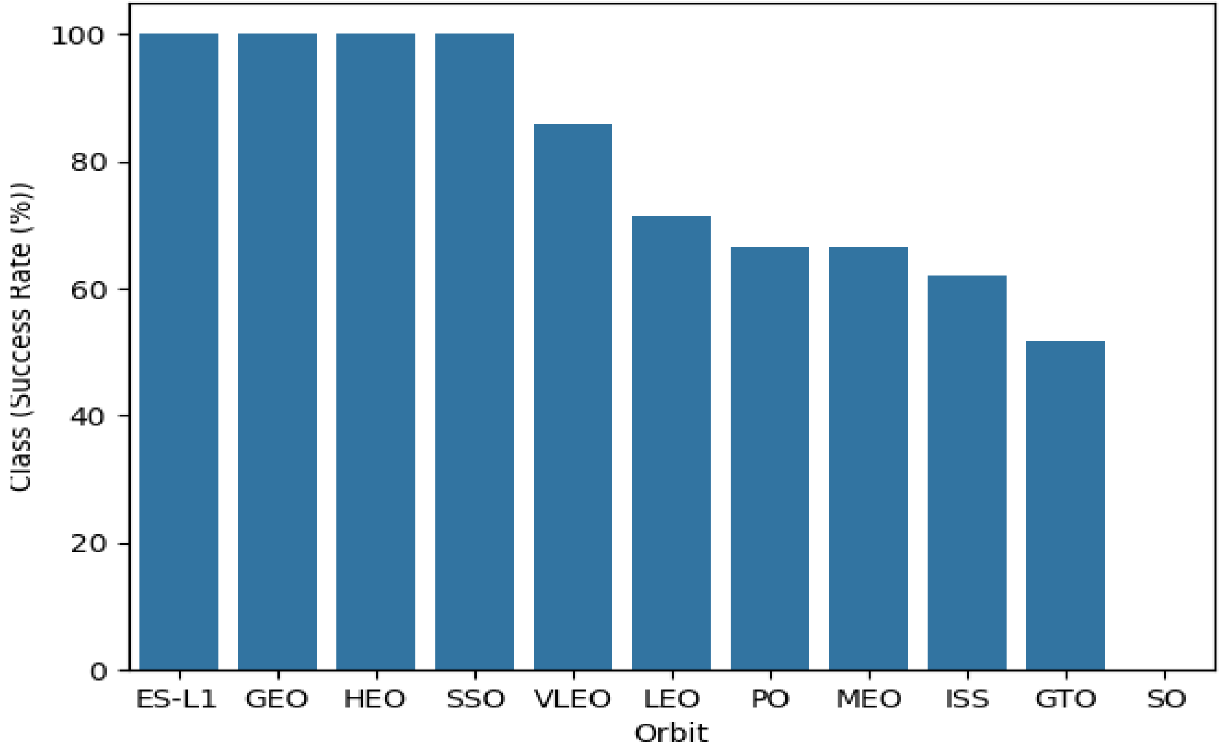
- Show a scatter plot of Payload vs. Launch Site

- Show the screenshot of the scatter plot with explanations



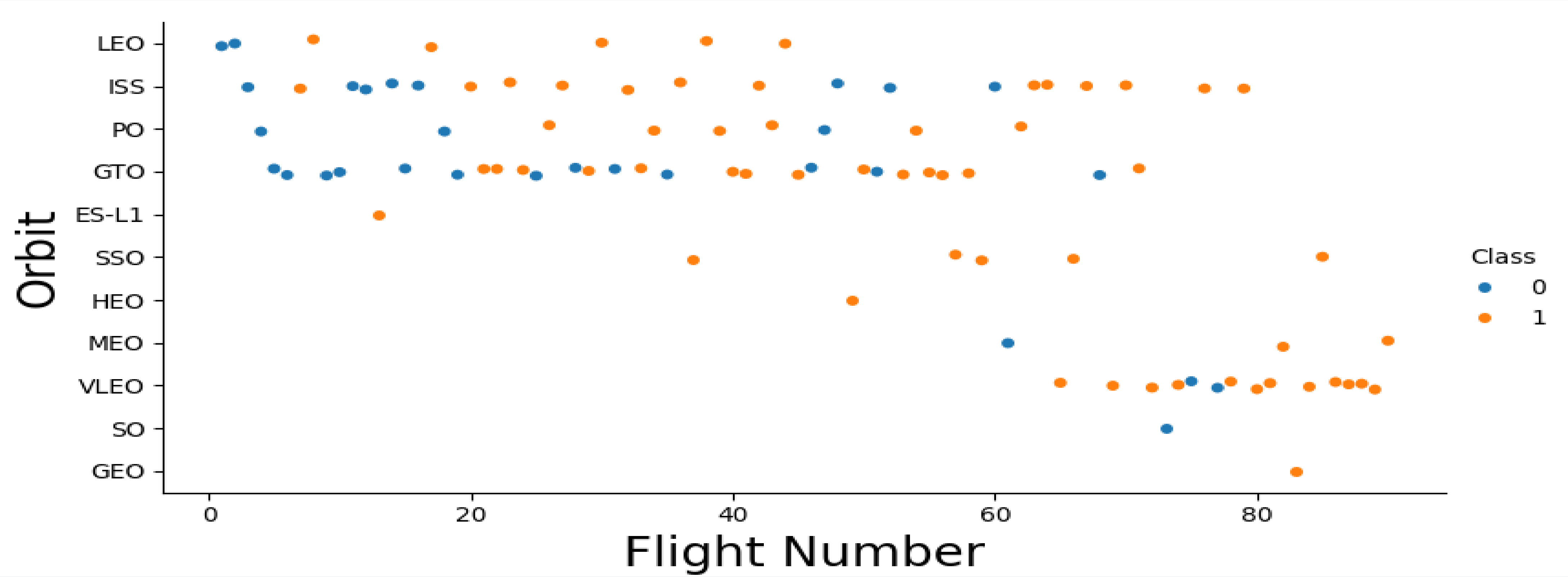
Success Rate vs. Orbit Type





Results shown in the right chart based on success scores show those with only one (successful) launch compared to GTO with 27 launches in the left chart. The outcome 'None None' is treated as a failure but it was a management decision to expend the booster. It was common on GTO orbits for boosters that reached their life limit to burn additional fuel reaching orbit and then drop for the final plunge into the ocean.

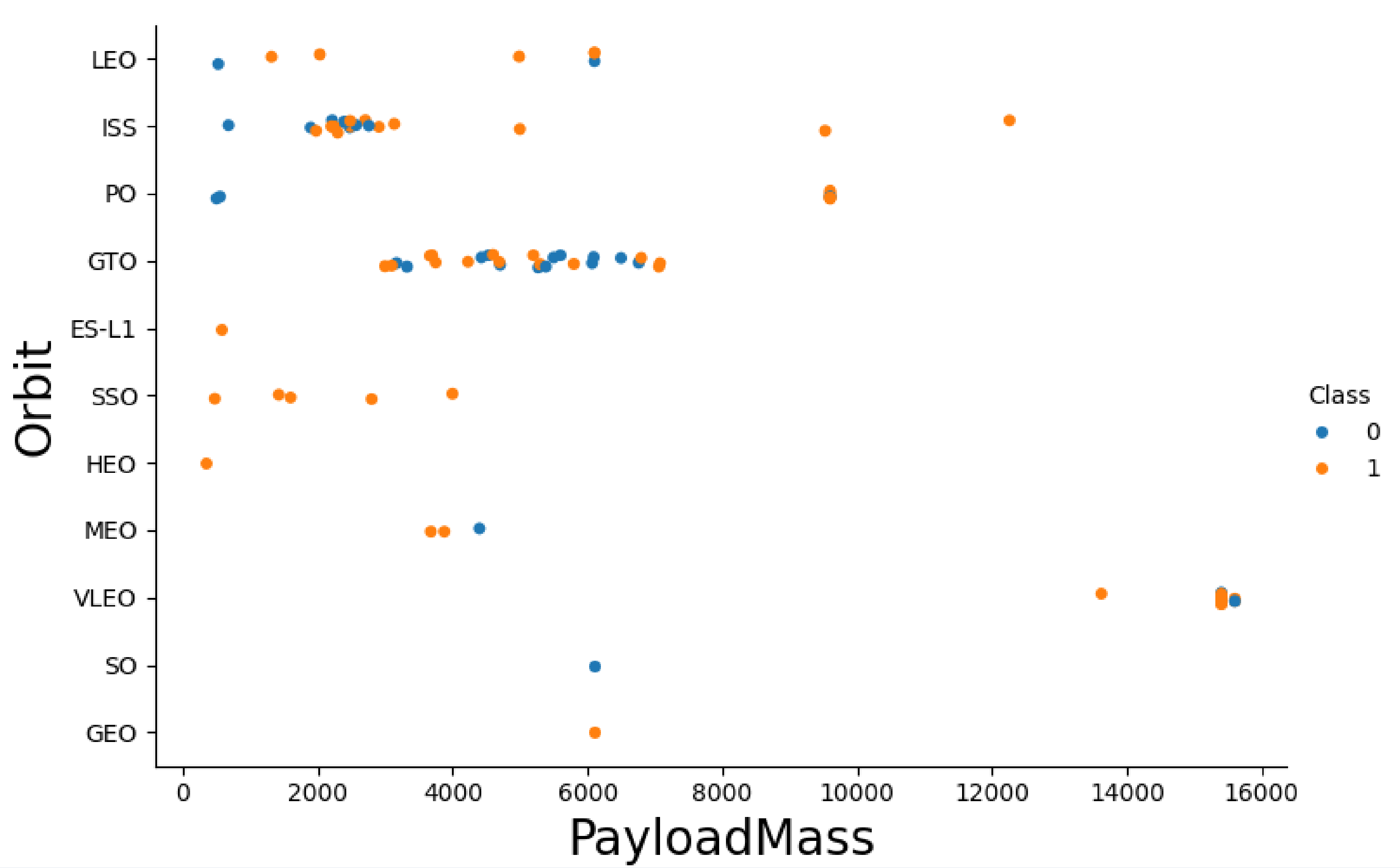
Flight Number vs. Orbit Type



New technology has improved rocket performance and brought in new commercial customers for satellite services in telephone , television , photographic, GPS and more. Much of this happening in the Very Low Earth Orbit region

Payload vs. Orbit Type

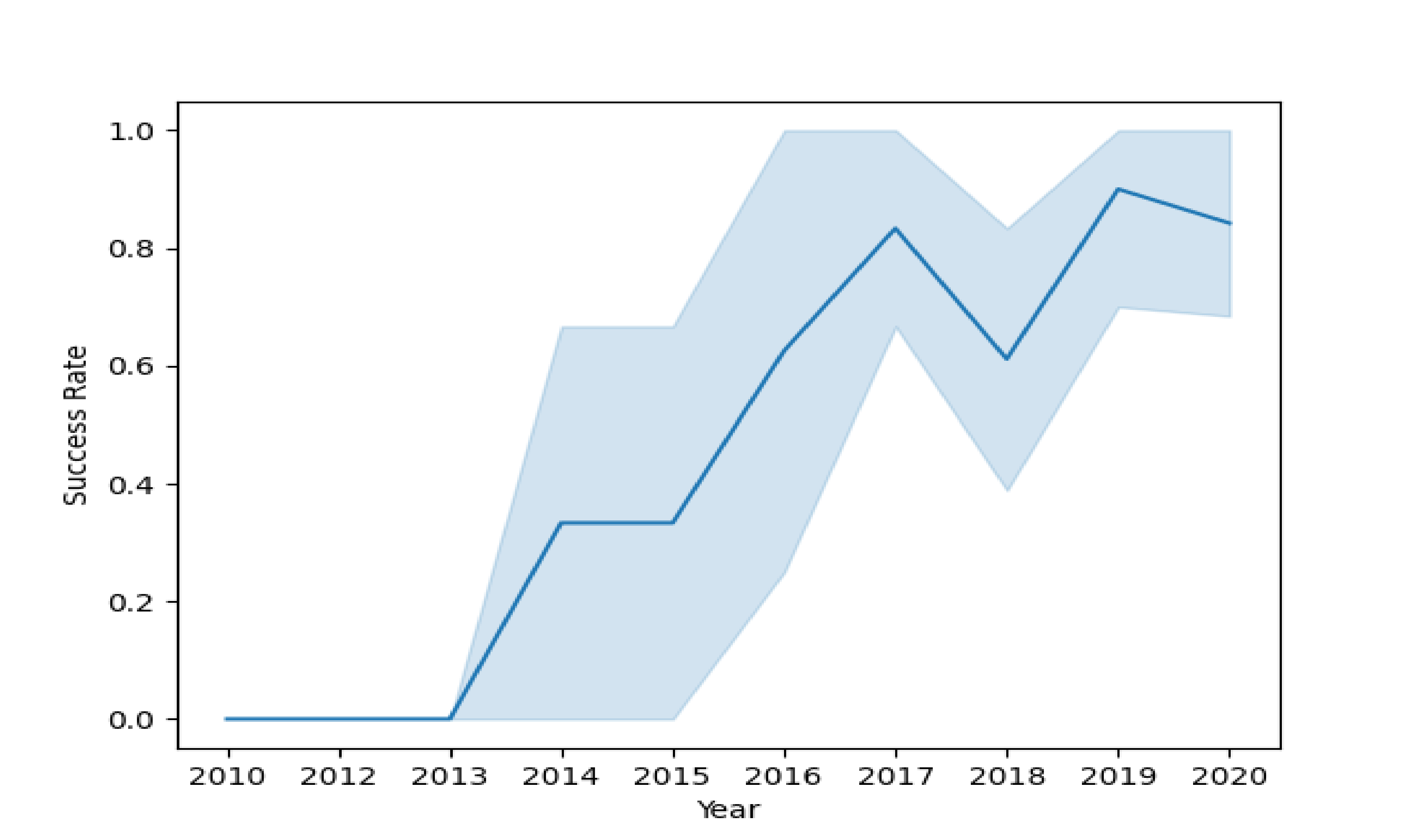
It appears launch payload continues to increase. In March, 2022 SpaceX launched a payload of 53 satellites weighing 308kg each (16,218kg). Similar deliveries have been completed as the graph shows within very Low Earth Orbit. However, typical payload remains in the 3000 to 7000 range.



Launch Success Yearly Trend

- Show a line chart of y
  average success rate

- Show the screenshot c
  scatter plot with
  explanations



All Launch Site Names

Task 1

Display the names of the unique launch sites in the space mission

```
%sql SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE;
```

✓ [11] 43ms

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Wrote an SQL satement to get one record for each unique (DISTINCT) Launch Site identifier

Launch Site Names Begin with 'CCA'

Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
%sql SELECT "Launch_Site" FROM SPACEXTABLE WHERE "Launch_Site"  LIKE 'CCA%' LIMIT 5;
```

✓ [12] 50ms

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
CCAFS LC-40
CCAFS LC-40
CCAFS LC-40
CCAFS LC-40

SQL Statement will select the first 5 records that start with 'CCA'

Total Payload Mass

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
%sql SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE CUSTOMER = 'NASA (CRS)';
```

✓ [13] 36ms

```
 * sqlite:///my_data1.db
Done.
```

SUM(PAYLOAD_MASS__KG_)
45596

SQL Statement will add total payload weight for Customer “NASA (CRS)”

Average Payload Mass by F9 v1.1

Task 4

Display average payload mass carried by booster version F9 v1.1

```
%sql SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1';
[23] 97ms
 * sqlite:///my_data1.db
Done.
```

AVG(PAYLOAD_MASS__KG_)
2928.4

SQL Statement to find average payload mass for Booster Version F9 v1.1

First Successful Ground Landing Date

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint:Use min function

```
%sql SELECT MIN("Date") FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)'
```

✓ [15] 42ms

```
 * sqlite:///my_data1.db
Done.
```

MIN(Date)
2015-12-22

SQL to find the date of the first successful landing outcome on ground pad

Successful Drone Ship Landing with Payload between 4000 and 6000

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
%sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE (PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000) AND ("Landing_Outcome" = 'Success (drone ship)');
```

✓ [16] 40ms

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

SQL for the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

Total Number of Successful and Failure Mission Outcomes

Task 7

List the total number of successful and failure mission outcomes

```
%sql SELECT COUNT("Mission_Outcome") FROM SPACEXTABLE GROUP BY "Mission_Outcome";
```

✓ [17] 42ms

```
 * sqlite:///my_data1.db
Done.
```

COUNT(Mission_Outcome)
1
98
1
1

Calculate the total number of successful and failure mission outcomes

Boosters Carried Maximum Payload

Task 8

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
%sql SELECT "Booster_Version" FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_=(SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
✓ [18] 51ms

 * sqlite:///my_data1.db
Done.
```

Booster_Version	
F9 B5 B1048.4	F9 B5 B1049.5
F9 B5 B1049.4	F9 B5 B1060.2
F9 B5 B1051.3	F9 B5 B1058.3
F9 B5 B1056.4	F9 B5 B1051.6
F9 B5 B1048.5	F9 B5 B1060.3
F9 B5 B1051.4	F9 B5 B1049.7

List the names of the booster which have carried the maximum payload mass

Note the boosters being reused: B1048.4, B1048.5, B1051.3, B1051.4, B1051.6, etc.

2015 Launch Records

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
%sql SELECT CASE SUBSTR("Date", 6, 2)WHEN '01' THEN 'January'WHEN '02' THEN 'February'WHEN '03' THEN 'March'WHEN '04' THEN 'April'WHEN '05' THEN
'May'WHEN '06' THEN 'June'WHEN '07' THEN 'July'WHEN '08' THEN 'August'WHEN '09' THEN 'September'WHEN '10' THEN 'Octber'WHEN '11' THEN 'November'WHEN
'12' THEN 'December'ELSE 'Unknown'  END AS month,"Landing_Outcome"='Failure (drone ship)',"Booster_Version","Launch_Site" FROM SPACEXTABLE WHERE
SUBSTR("Date", 0, 5) = '2015';
```

✓ [19] 43ms

```
 * sqlite:///my_data1.db
Done.
```

month	Landing_Outcome='Failure (drone ship)'	Booster_Version	Launch_Site
January	1	F9 v1.1 B1012	CCAFS LC-40
February	0	F9 v1.1 B1013	CCAFS LC-40
March	0	F9 v1.1 B1014	CCAFS LC-40
April	1	F9 v1.1 B1015	CCAFS LC-40
April	0	F9 v1.1 B1016	CCAFS LC-40
June	0	F9 v1.1 B1018	CCAFS LC-40
December	0	F9 FT B1019	CCAFS LC-40

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
%sql SELECT "Landing_Outcome",COUNT("Landing_Outcome") FROM SPACEXTABLE WHERE "Date" BETWEEN '2010-06-04' and '2017-03-20' GROUP BY "Landing_Outcome"
ORDER BY 2 DESC
```

✓ [20] 43ms

```
 * sqlite:///my_data1.db
Done.
```

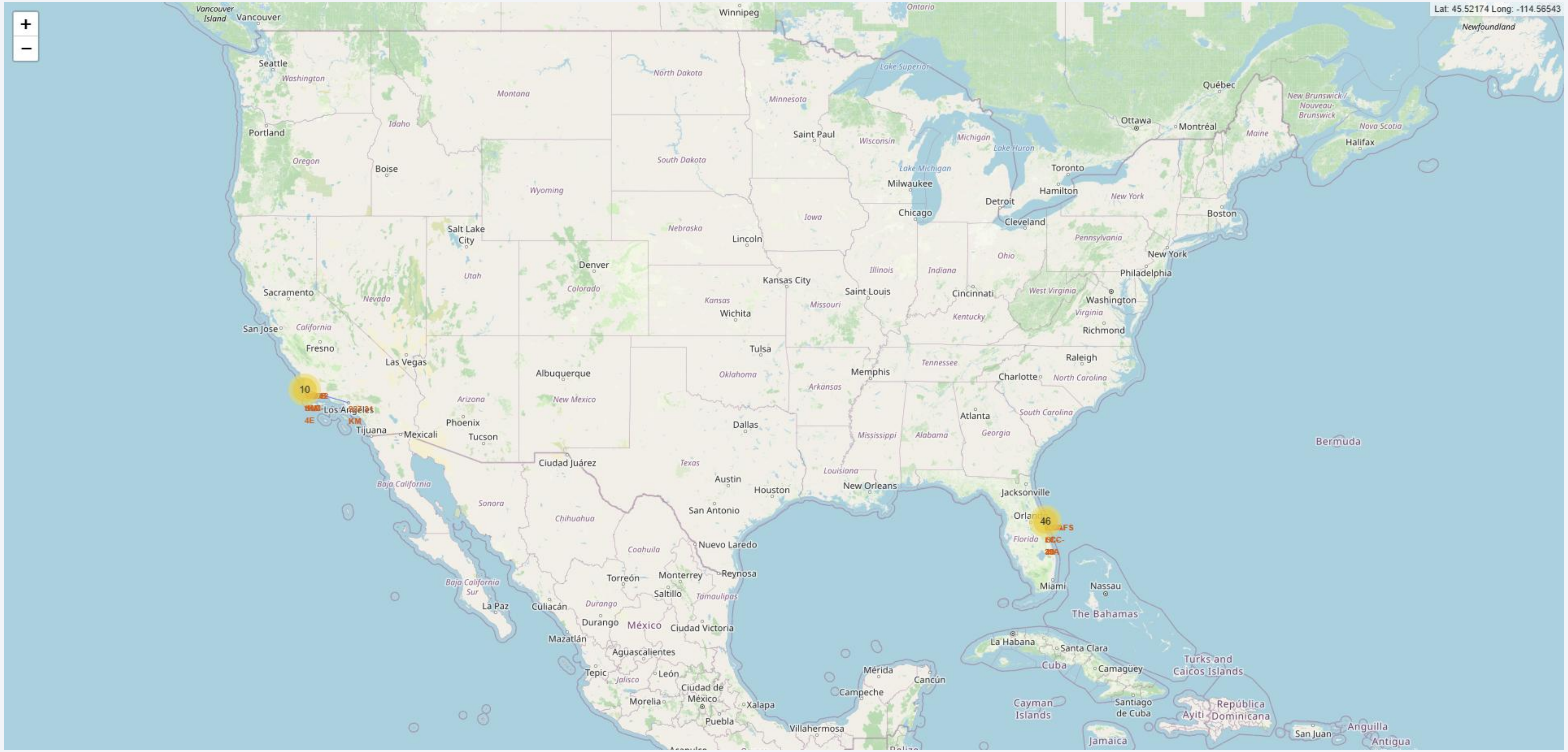
Landing_Outcome	COUNT(Landing_Outcome)
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

Section 3

Launch Sites Proximities Analysis

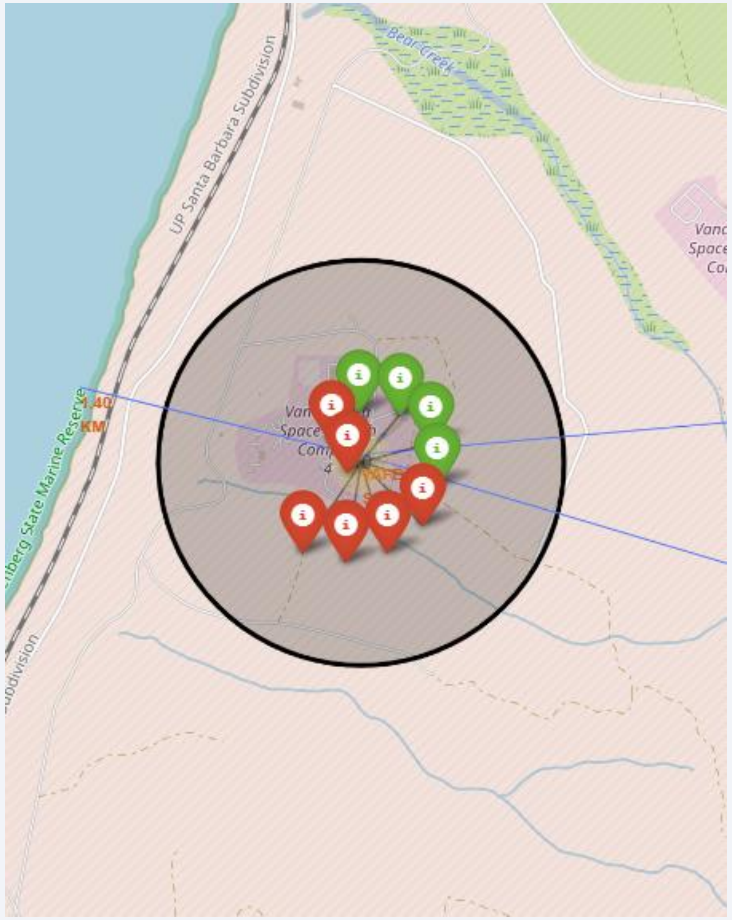
All Launch Site Locations on a Global Map Screenshot



Launch sites are displayed at Vandenburg SFB and Cape Canaveral AFS

Color-Labeled Folium Map Landing Pad Outcomes

- Marker Clusters may be opened to display outcomes of rockets launched from the same latitude and longitude.
- Green markers indicate a successful recovery, red markers indicate failure.

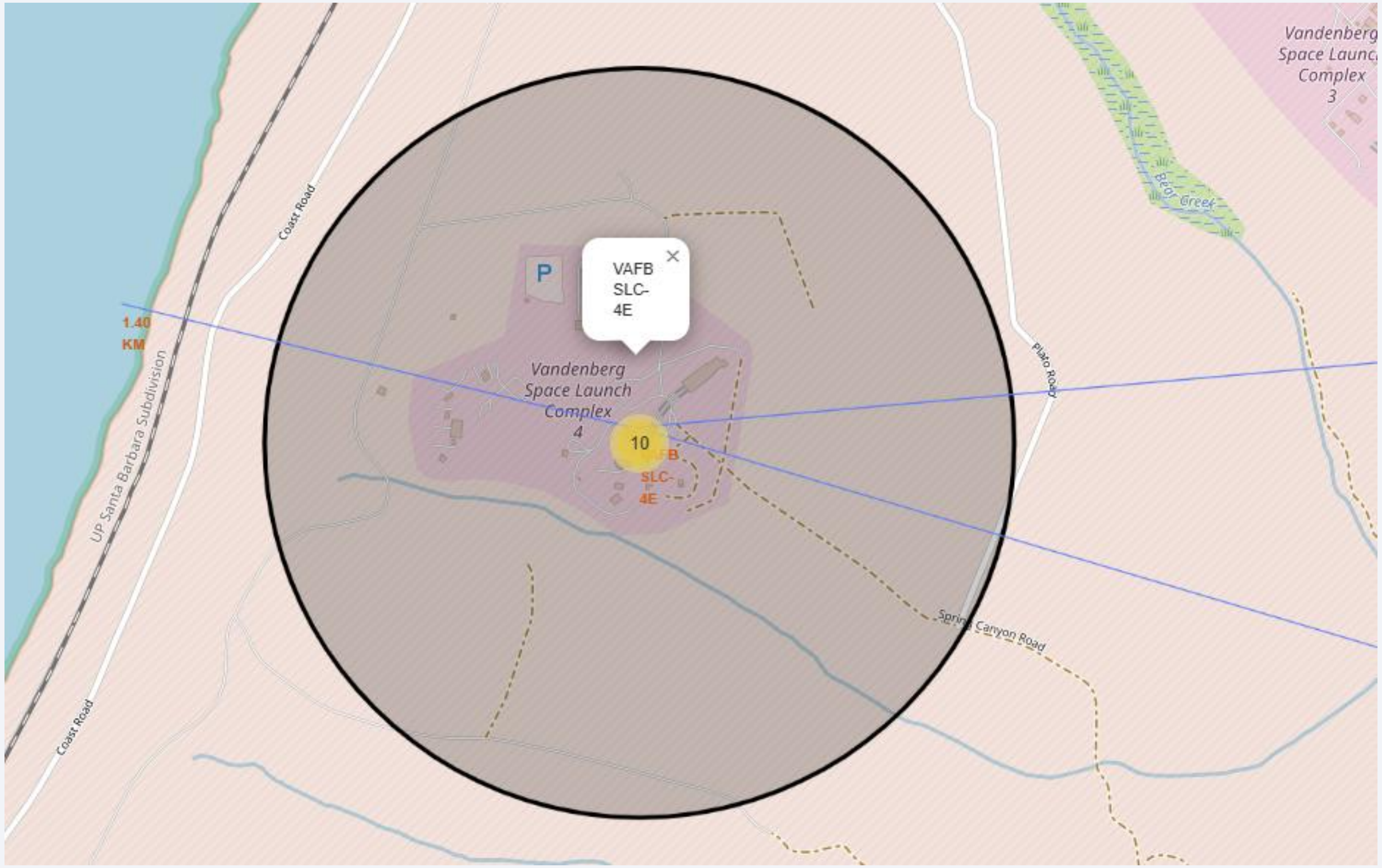


VAFB SLC-4E

KSC LC-39A

Launch Site Folium Map Geographical Relations

- Polylines allow you to measure distances from roads, railroads, coastlines, towns, etc.
- This shows the distance to the coastline (1.4 Km), Lompoc, the nearest small town (14.02 Km), and Los Angeles (227.31 Km)
- Proximity to the coast is good, population density is low, and transportation sources are available. Launches can be southerly and westerly. No launches may be to the west

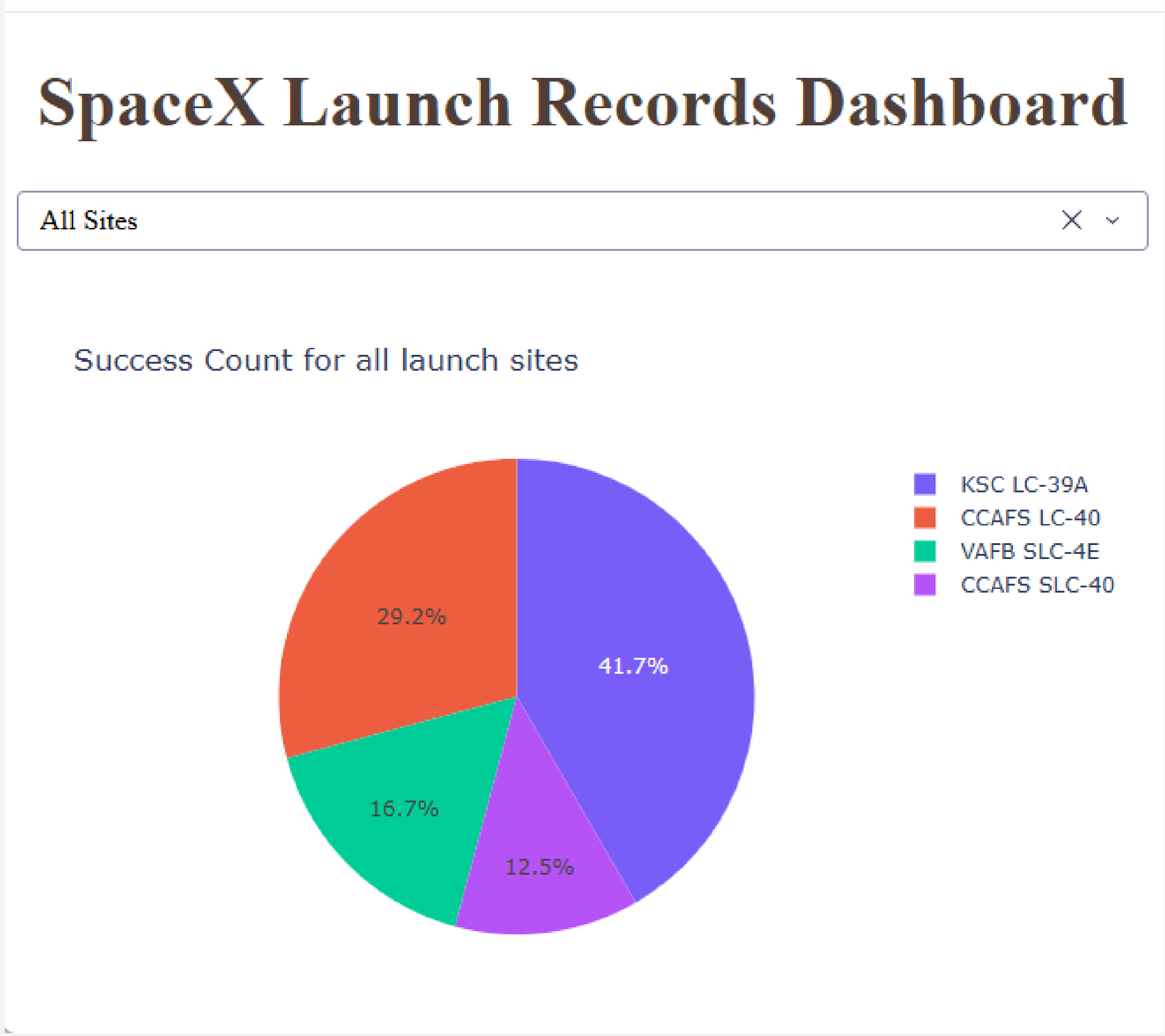


Section 4

Build a Dashboard with Plotly Dash

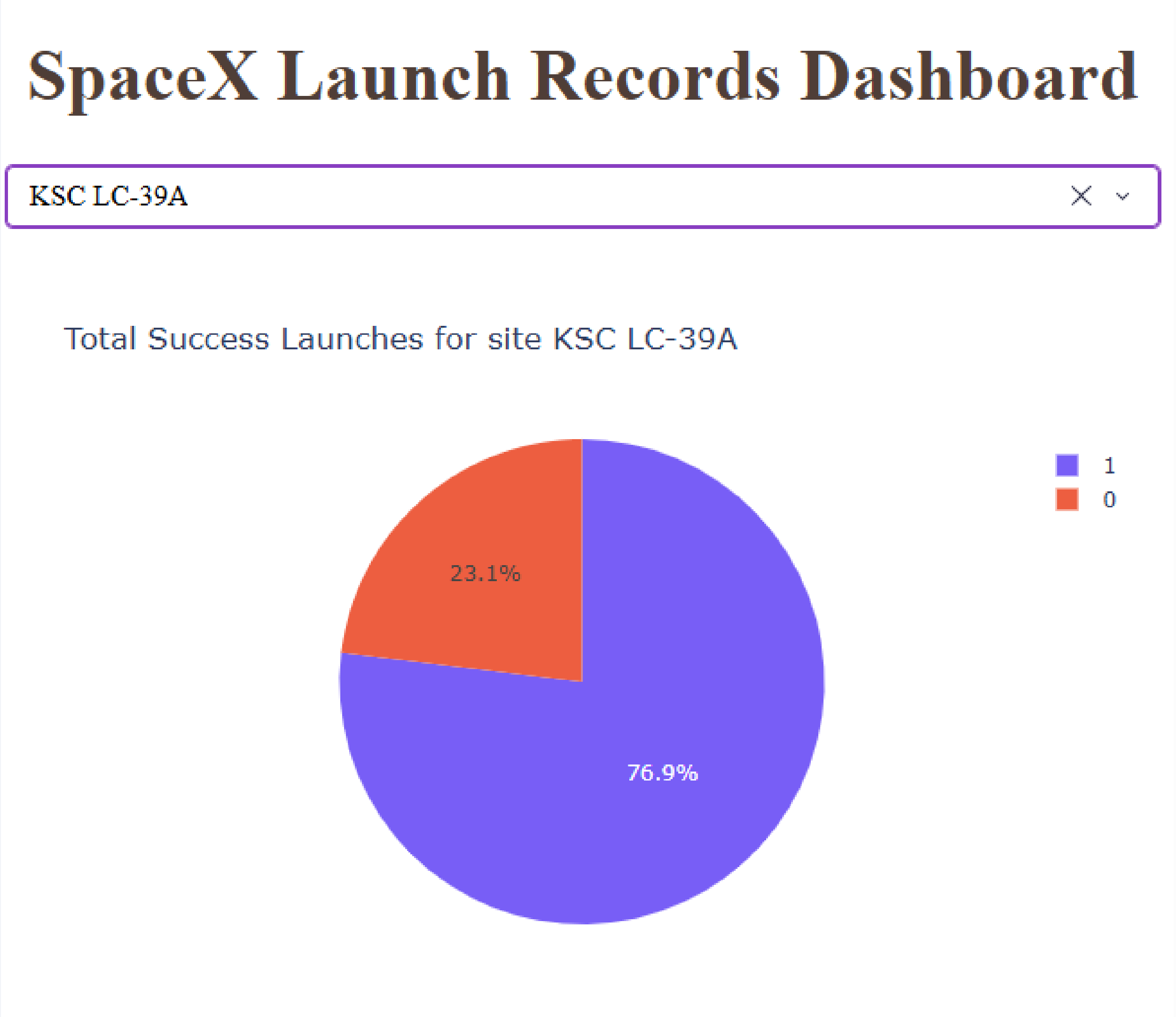
Dashboard Launch Site Performance

- Launch Site KSC LC-39A was the most successful

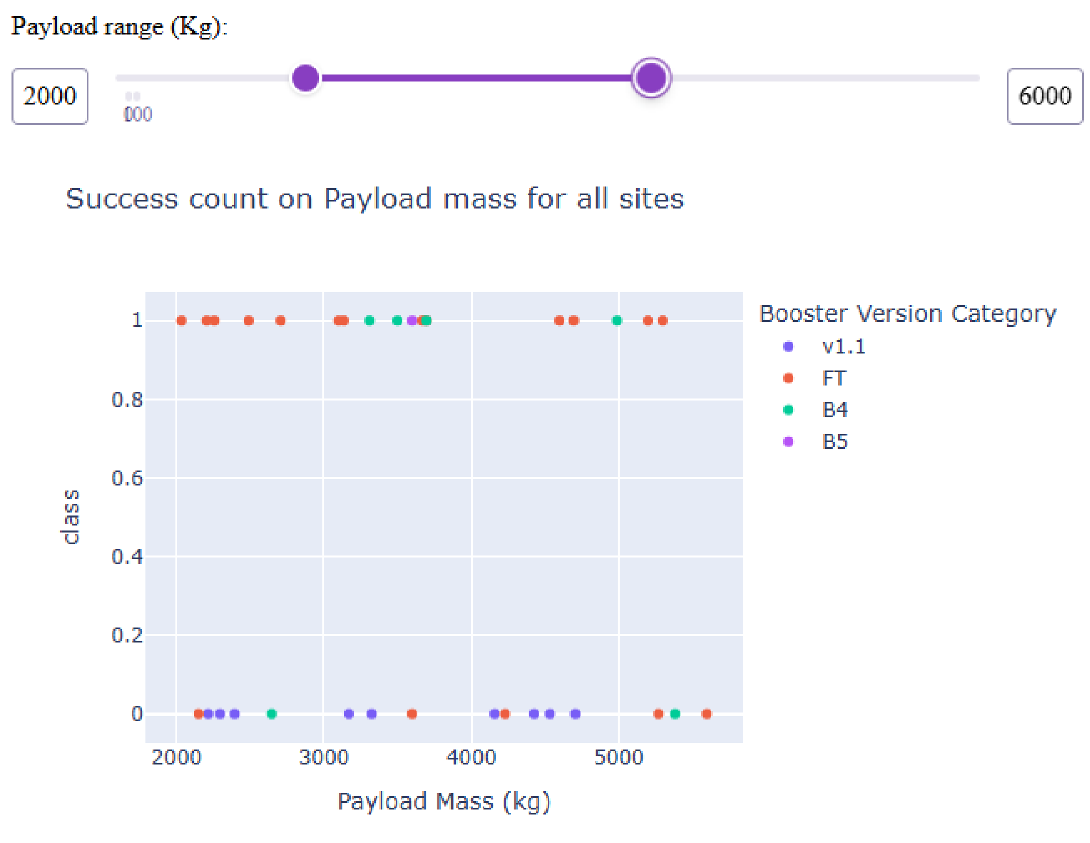


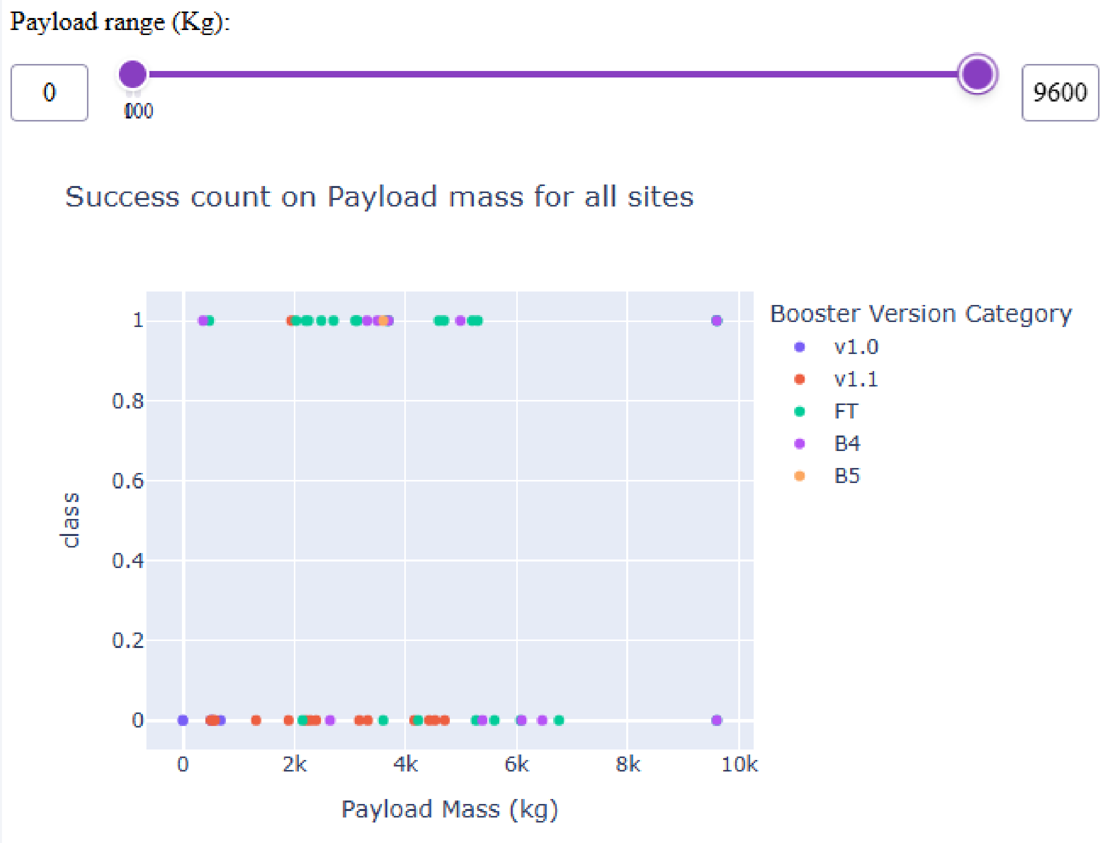
Launch Site KSC LC-39A

- This shows that Launch Site KSC LC-39A was successful 76.9% of the time



Booster Performance Advancement





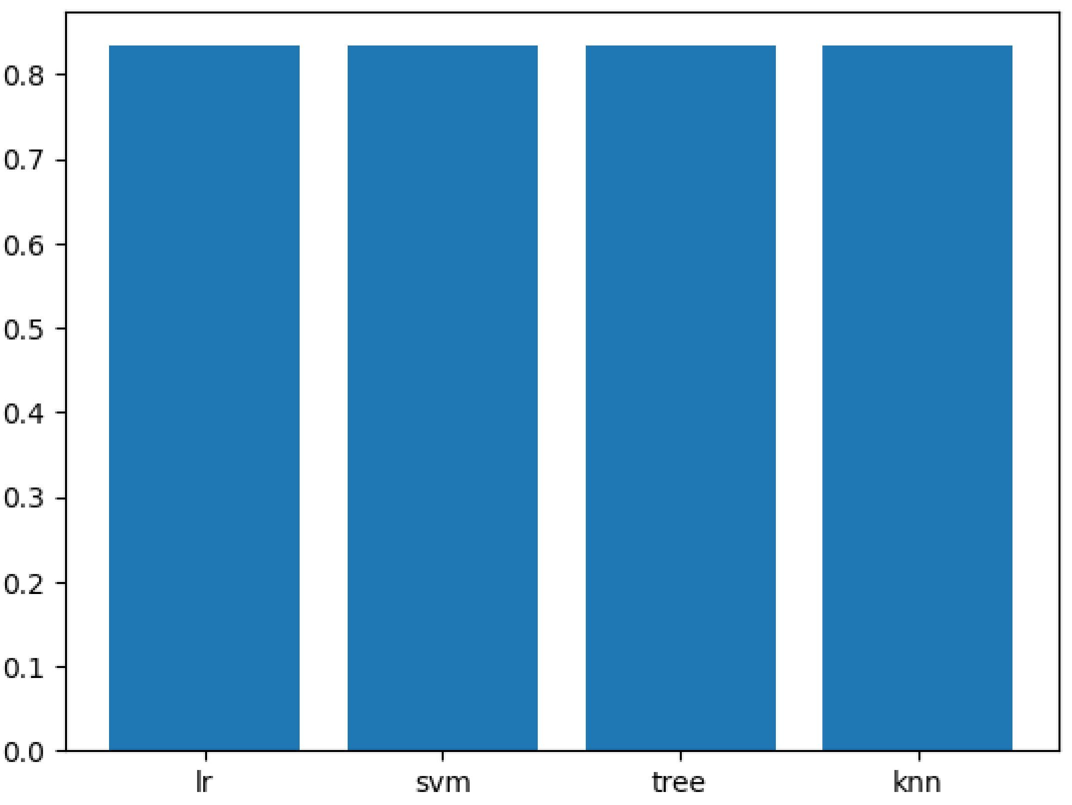
- Most of the activity appears between 2000Kg and 6000Kg. The Falcon 9 V1.1 and V1.0 were not very successful in returning .The Falcon 9 FT seemed to start to hold enough fuel to return for landing. Block 4 and Block 5 solved the 30% return fuel need. It still remains higher and faster separation requires more fuel burn.

Section 5

Predictive Analysis (Classification)

Classification Accuracy

- The built model accuracy for all built classification models, in a bar chart

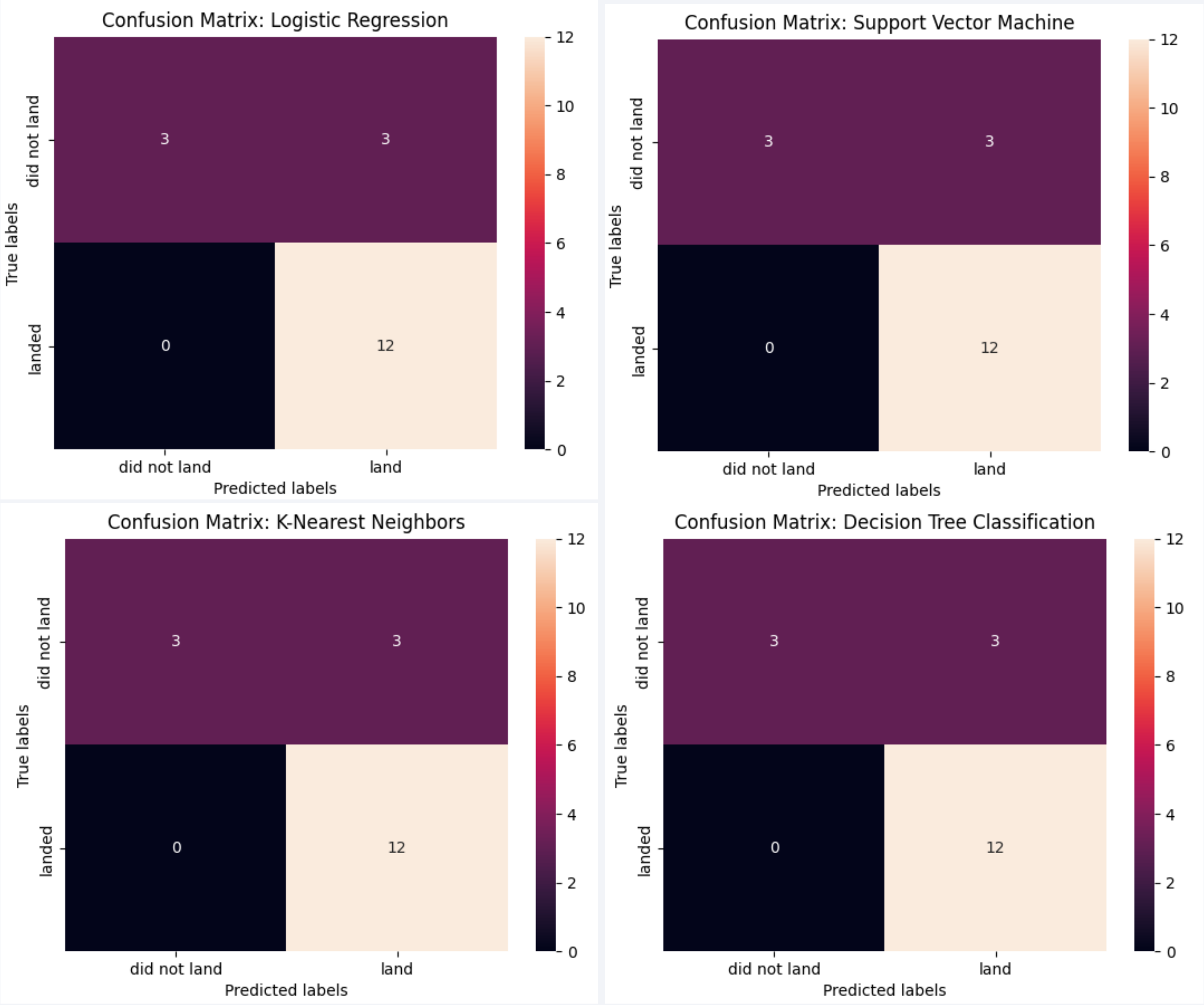



```
  Method                            % Accuracy
Logistics Regression (lr) :  0.8333333333333334
Support Vector Machine (svm) :  0.8333333333333334
Decision Tree (tree) :  0.8333333333333334
K-Nearest Neighbors (knn) :  0.8333333333333334
```

Confusion Matrix

'Support Vector Machine (svm)', 0.8333333333333334] has a slight but not statistically significant edge.

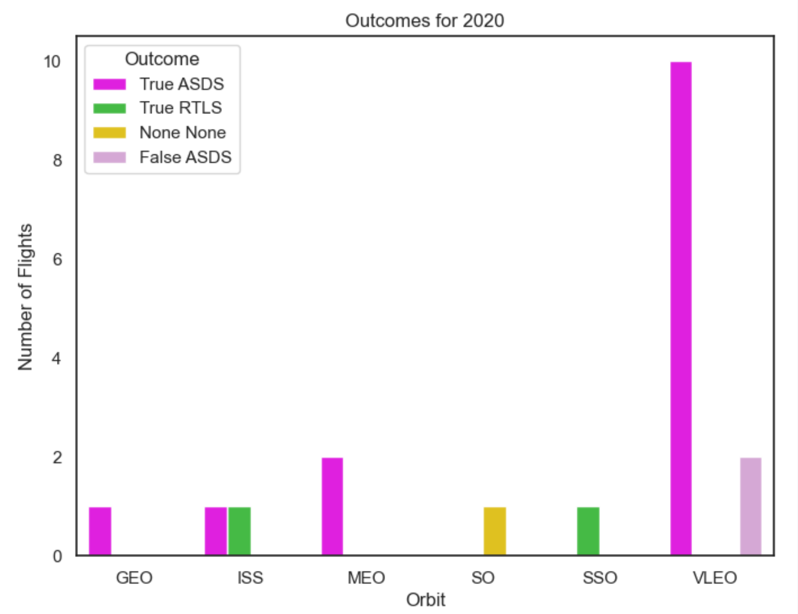
All four tests had an identical Confusion Matrix. The confusion matrix shows there were 3 true positives, 3 false negatives, 0 false positives, and 12 true positives

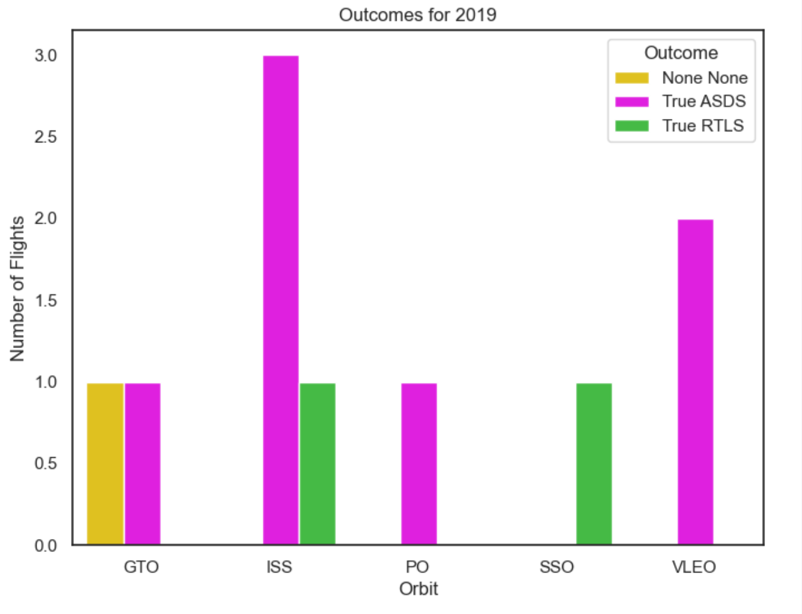


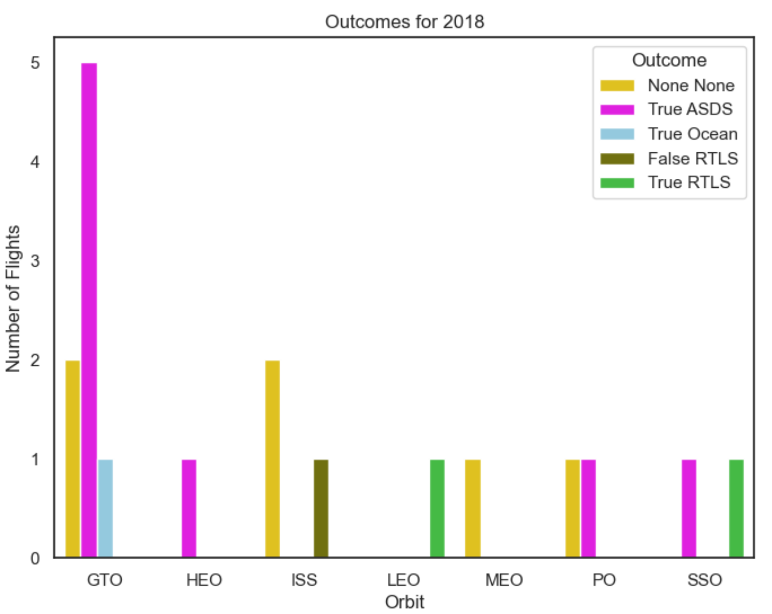
Conclusions

- VLEO became the busiest orbit in 2020 (Page 46)
- The “None None” outcomes decreased as the allowed flights of a Falcon Block 5 booster increased from 10, 20, 30 flights starting in the middle of 2018 (Page 47)
- Drone ship landings allow for higher performance, heavier payloads, and more efficient, less fuel-intensive trajectories. Ground pad landings are much cheaper (Page 47)

Appendix 1

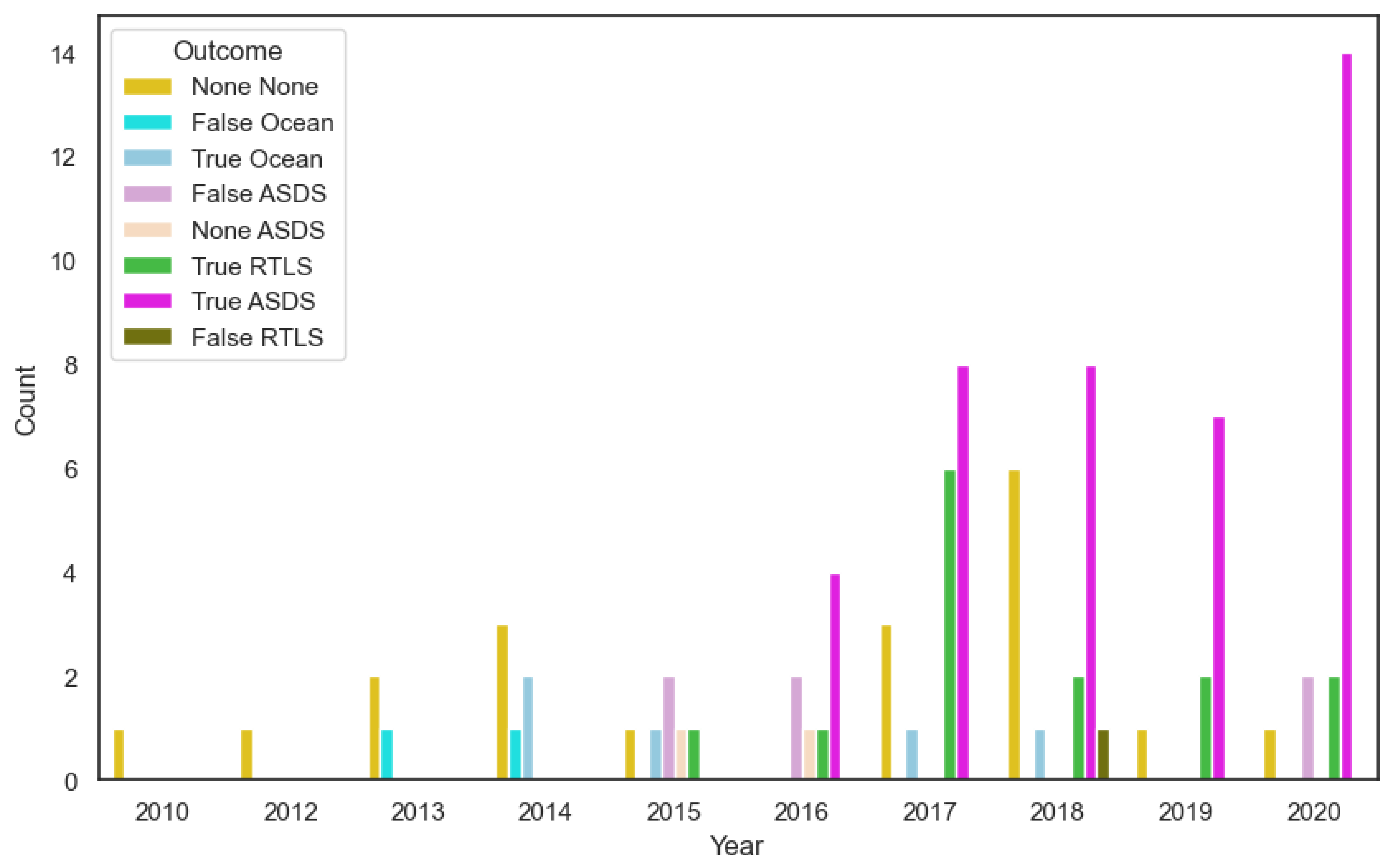






VLEO became the busiest orbit in 2020 after beginning in 2019. This is for StarLink and competitors

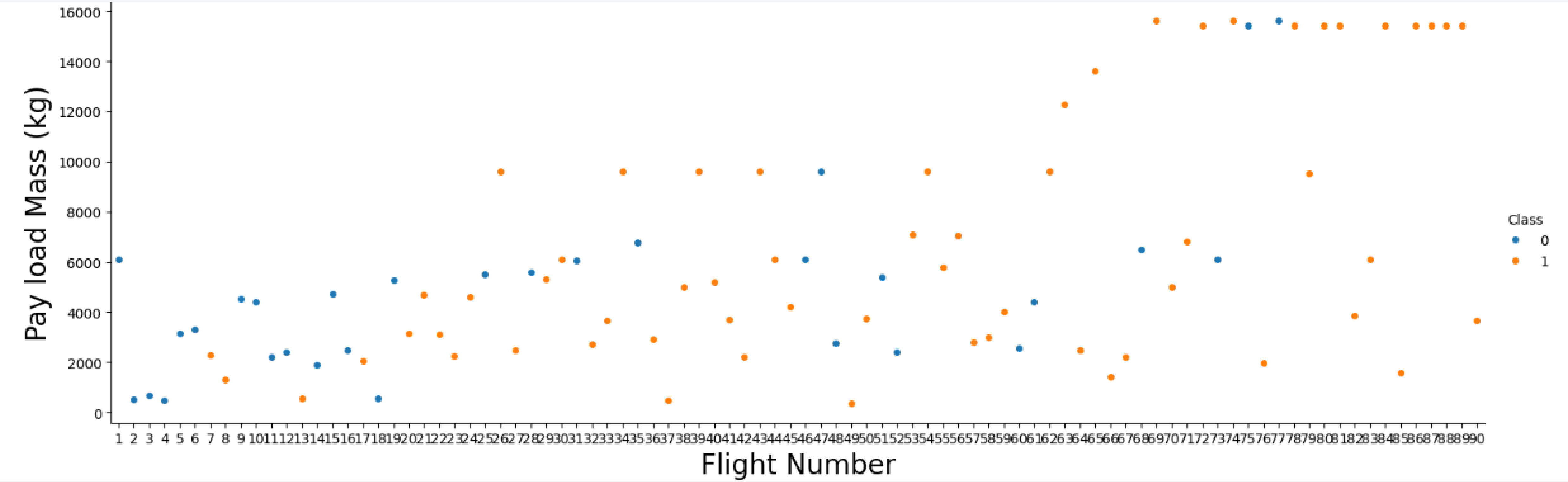
Appendix 2



The "None None" outcomes decreased as the allowed flights of a Falcon Block 5 booster increased from 10,20,30 flights started in the middle of 2018.

Drone ship landings are more frequently chosen because they allow for higher performance, heavier payloads, and more efficient, less fuel-intensive trajectories. Ground pad landings are much cheaper

Chart: FlightNumber, Payload and the Outcome




```
sns.catplot(y="PayloadMass", x="FlightNumber", hue="Class", data=df, aspect = 3)
plt.xlabel("Flight Number",fontsize=20)
plt.ylabel("Pay load Mass (kg)",fontsize=20)
plt.show()
```


Return

Thank you!